SUPPLY CHAIN ANALYTICS • EDA SUMMARY

Supply Chain EDA Insights

Global Supply Chain EDA – Key Insights & Strategy

Performance • Disruptions • Risk–Cost • Forecast • Actions

Analysis scope (dataset snapshot)

- **10,000 shipments** analyzed (end-to-end delivery performance)
- Global origins & destinations (major trade hubs)
- Multi-modal logistics: **Sea** (primary) and **Air** (express segment)

Key questions addressed

- Where are the highest-volume corridors and bottlenecks?
- Which disruption events drive frequency vs. severity of delays?
- How does geopolitical risk translate into cost premium?

Executive Summary

Snapshot of reliability, modal mix, key hubs, and delay pattern (N = 10,000 shipments)

ON-TIME DELIVERY (OTD)

87.1%

8,710 on-time • 1,290 late

HIGH-DELAY RATE

8.8%

Delay > 3 days (focus segment)

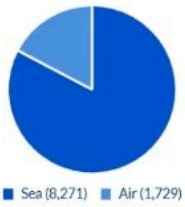
DATA QUALITY

8,733

Missing Disruption_Event

Transport Mix (Orders)

Sea dominates volume; Air serves express segment

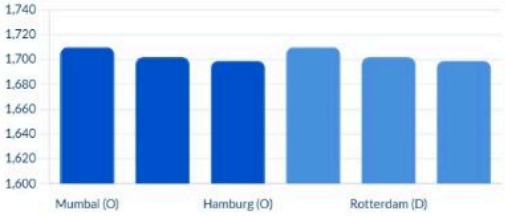


Sea: 8,271 (~82%)

Air: 1,729 (~17%)

Top Trade Hubs (Orders)

Balanced export origins; concentrated import destinations



Top origins: Mumbai (1,710) • Shenzhen (1,702) • Hamburg (1,699)

Top imports: Felixstowe (1,710) • Rotterdam (1,702) • New York (1,699)

Delay Profile

Most within 0–2.5 days

~90% fall within **0–2.5 days** delay

Long tail extends to **20–50 days** (rare)

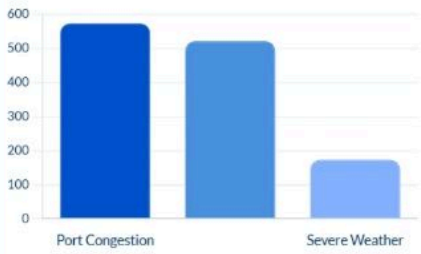
Disruption Analysis

Frequency vs. severity of disruptions and current mitigation response (known disruption events)

Top disruption drivers (≈86% of known delays): Port Congestion (573) • Geopolitical Conflict / Route Diversions (521) • Severe Weather (173)
Conflict events are less frequent than congestion but **most severe** (longest delays and high-delay concentration).

Primary Causes (Count)

Known disruption events linked to delays

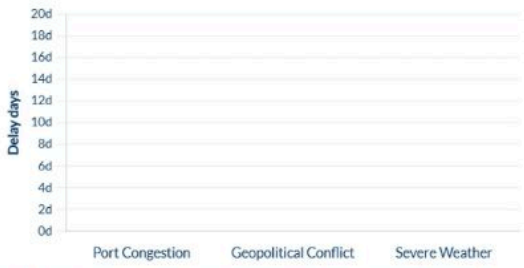


Operational driver: congestion is frequent but typically short (2–4 days).

Strategic driver: conflict creates longer diversions (10–16 days; outliers near 20).

Severity (Delay Days)

Distribution proxy (box + whiskers) by disruption type



Conflict has the highest median and widest spread (systemic severity).

Congestion is consistent (predictable nuisance) while **weather** sits mid-range.

Mitigation

Observed response choices

Standard: 9,283
(~93% of orders)

Re-route: 368
Used sparingly

Expedite Air: 349
Reserved for critical loads

Risk & Cost Analysis

Route risk mix, risk premium on shipping cost, and disruption-related financial exposure

Cost increases with geopolitical risk: High-risk average cost **$13,834** vs Low-risk **$8,966** (**+54%**).
Risk distribution is centered on Medium risk (50%), with ~24% of volume operating in High-risk zones.

Risk Mix (Orders)

Share of shipments by geopolitical risk category

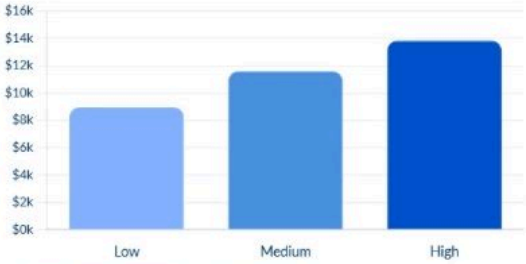
■ Medium (50.0%) ■ Low (26.2%) ■ High (23.7%)

Medium risk: 5,004 (50.0%) is the operating core.

High risk: 2,374 (23.7%) drives exposure to conflict-driven disruptions.

Risk Premium on Cost (Avg $)

Average shipping cost rises with risk level (+~$2.4k per step)



High vs Low: +$4,868 per shipment (+54%).

Confidence bands are not shown here; interpret as **directional** risk-based pricing.

Spend Focus

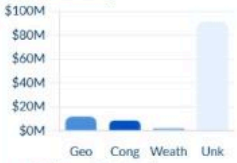
High-cost segment & disruption spend

HIGH-COST SHIPMENTS

10%

1,000 of 10,000

Disruption Spend (Known vs Unknown)



"Unknown" disruptions account for **$91.3M** — a data governance gap.

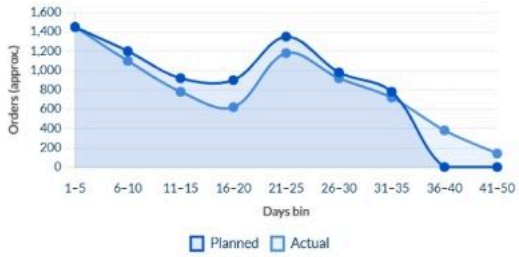
Performance & Forecast

Planned vs actual delivery timing, transport delay parity, seasonality, and 4-month delay outlook

Two operational "lanes" appear end-to-end: Express (≤5 days) and Standard (≈25–35 days). Actuals show a **long-tail beyond 35 days** (up to ~50 days in rare cases).
Short-term outlook (Feb–May 2026) is stable around **0.95 days** average delay (low variance).

Planned vs Actual Delivery (Distribution)

Express peak at ≤5d; standard peak around 25–35d; actual extends further



Interpretation: Express lane aligns with Air freight; Standard lane aligns with Sea.

Actual tail suggests **variability in long-haul legs** (ports, routing, congestion).

Seasonality & 4-Month Outlook

Aug 2025 peak (~1.3d) followed by stabilization; forecast stays ~0.95d

Peak period: Aug 2025 (~1.3d avg delay).

Forecast: Feb–May 2026 ≈ 0.95d (low variance).

Mode Parity

Avg delay by transport mode

SEA (AVG) **0.95d**

AIR (AVG) **0.92d**

Difference

0.03d
≈ 45 minutes

Implication: Air reduces average delay only marginally vs Sea.

Choose Air primarily for **lead-time targets** (not delay reduction alone).

Actionable Recommendations

Prioritized actions to improve reliability, reduce high-delay exposure, and close data governance gaps

Next 2 quarters — KPI targets

Focus on high-delay reduction and better disruption classification to improve cost control

OTD	HIGH-DELAY (>3d)
87% → 90%	**8.8% → 6.0%**

1) Route diversification for critical corridors

Establish Plan-B ports/lanes for **Mumbai → Felixstowe / Rotterdam** to protect high-volume flow and reduce port bottleneck exposure.

Target: **-25%** major delays to Felixstowe/Rotterdam

2) Active mitigation triggers (standard → reroute/expedite)

Auto-escalate when **Conflict/Congestion** is flagged; use **Air** for high-value SKUs and **re-routing** for blocked corridors.

Guideline: expedite only when cost of delay exceeds "urgency tax"

3) Contract & cost audit (focus on the vital few)

Audit premium carriers and service levels—EDA suggests **higher shipping cost does not guarantee fewer delay days.**

Prioritize the **top 10%** high-cost shipments for negotiation & policy controls

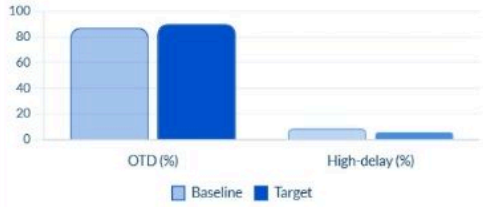
4) Risk & data governance improvements

Improve disruption tagging at origin/destination handoffs; reduce "Unknown" event classification to support targeted mitigation.

Target: reduce **Unknown disruptions by 50%**

Targeted KPI uplift

Baseline vs target (next 2 quarters)



Implementation checklist

- Define escalation rules (conflict/congestion thresholds)
- Set corridor playbooks (Plan-B ports and carriers)
- Improve disruption tagging (reduce missing/unknown fields)